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A TECHNICAL SEMINAR REPORT ON

**UNRAVELING WITH AI DECISION: INTERPRETING
AND BREAKING DOWN THE BLACK BOX USING SHAP**

SUBMITTED TO
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (AI & DS)
PIMPRI CHINCHWAD UNIVERSITY

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ABSTRACT

Explainable AI, or XAI, is a growing field in modern machine learning because it focuses on building models more transparent, easy to understand, and more reliable. Many machine learning models operate like “black boxes”: they give accurate results, but it is difficult to know exactly how they arrived at those results. This lack of understanding can be a big problem in areas where decisions are very important, such as medical diagnosis, financial decisions, and security. Basically, XAI helps us understand why a model made a particular analysis.

The two most used techniques in XAI are SHapley additive explanations and interpretable local model agnostic explanations. Both methods aim to explain the model predictions by showing how much each input feature contributes to the result. This report covers the basics of SHAP and LIME, focusing on how they work and how they handle challenges such as feature correlation and model dependency. These factors often affect the quality and reliability of the explanations, it is important to take in account when choosing the appropriate technique.

The report also explores the strengths and weaknesses of SHAP and LIME. While they are powerful tools, they are not perfect and their explanations can be misleading if not used properly. To show how these techniques are applied in real situations, we analyze a biomedical case study where the interpretability of the model is essential. The real-world examples and solutions included in the report demonstrate how clear explanations help make better decisions in high-risk environments.

Overall, this discussion shows that model explanations must be reliable. The techniques developed under XAI are very important in making machine learning more secure, transparent and reliable.

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Chapter 1 Introduction

Modern machine learning and deep learning models are incredibly powerful, but often behave like "black boxes," where their decisions are difficult to interpret. This is a great concern in sensitive areas such as healthcare and finance, where there is a need for transparency.

Explainable AI, XAI, makes such models more transparent, and thereby more trustworthy. There are two most common explainable AI techniques: SHAP and LIME. SHAP uses game-theory concepts to explain predictions, while LIME generates simple local models to showcase how features contribute to an individual prediction.

1.1 Overview

This survey investigates the role of Explainable AI in enhancing transparency and gaining confidence in complex machine learning models, mostly applied to sensitive domains such as healthcare or finance. This review focuses on two of the most widely used XAI methods: SHAP and LIME. Both explain predictions by providing feature importance. SHAP is based on game-theory ideas, yielding consistent explanations. LIME, on the other hand, relies on simple local models to explain individual predictions.

The study further looks into challenges related to feature collinearity and the impact of various models' on explanation stability. A biomedical case study shows, in practice, how SHAP and LIME can support better decision-making. The paper surveys their strengths, weaknesses, and when each method is most useful.

1.2 Motivation

Understandably, with powerful machine learning models becoming commonplace in sensitive areas such as health and finance, it is very important to understand their decisions. Black-box models, due to their lack of transparency, diminish trust and make critical decisions precarious. It is this aspect that motivates the use of Explainable AI approaches such as SHAP and LIME. Their study helps to understand how reliable the explanations provided by these approaches are and how they can be improved for real-world applications.

1.3 Problem Statement And Objectives

Most modern ML models are black boxes, and it is difficult to understand how features drive predictions. The lack of interpretability often becomes a serious issue in real high-stakes applications when features are correlated or different models provide inconsistent explanations.

- Examine how SHAP and LIME generate explanations of importance of characteristics.
- To study the influence of feature collinearity and model choice on explanation stability.
- Compare the strengths and limitations of SHAP and LIME.
- Assess these methods with actual biomedical data.
- Guide on choosing appropriate XAI techniques for sensitive applications

1.4 Scope

The scope of this study is to analyze and compare the interpretability of modern machine-learning models using two widely adopted Explainable AI techniques: SHAP and LIME. The report focuses on understanding how these methods generate explanations, how reliable and stable those explanations are, and how they behave under challenges such as feature collinearity and model dependency. The scope also includes evaluating SHAP and LIME on a real biomedical dataset to observe their practical usefulness in high-stakes domains. This study does not involve implementing new algorithms; instead, it aims to provide a structured understanding of when and how these XAI tools should be applied for transparent, reliable, and trustworthy decision-making.

1.5 Methodologies Of Problem Solving

1. Problem Identification:

Modern ML models act as black boxes, making it difficult to understand how and why they make predictions. This lack of interpretability becomes a major problem in sensitive domains, where reliable and transparent explanations are essential for trust and decision-making.

2. Information Gathering:

It involves collecting relevant research papers, datasets, background theory, and existing work on SHAP, LIME, and XAI. This step ensures a clear understanding of current methods, challenges, and gaps before applying or evaluating the techniques.

3. Analysis of Existing Techniques:

It involves studying how current XAI methods like SHAP and LIME generate explanations, along with their strengths, weaknesses, and limitations. This helps identify gaps such as instability, sensitivity to feature collinearity, and inconsistencies across different models.

4. Solution Approach:

The solution approach applies SHAP and LIME to the model and evaluates the explanations to understand feature influence and reliability.

5. Evaluation and Validation:

Evaluation and Validation confirm whether the generated SHAP and LIME explanations are accurate, consistent, and aligned with the model's true behavior.

6. Presentation and Interpretation:

Presentation and Interpretation involve clearly displaying the SHAP and LIME results through visuals and explanations so users can easily understand how the model makes decisions.

Chapter 2 Literature Survey

Introduction to Literature Review: This literature review summarizes key research done on explainable AI methods like SHAP, LIME, and other related works. It emphasizes how these techniques work, where they can be useful, and what limitations they still have. Based on the work that has already been done, we understand the current challenges in model interpretability and identify places where better, more reliable explanation methods are needed.

1) xxAI - “xxAI / From Visualizations Towards Actionable XAI”.

It provides a comprehensive overview of current XAI approaches, challenges, and interdisciplinary applications. This work outlines a large taxonomy of explanation methods that include local and global explainability. It also addresses practical concerns about the evaluation of explanations, handling uncertainty, and construction of trustworthy AI systems for deployment into healthcare and environmental science.

Key contributions:

- Broad taxonomy of XAI methods: local versus global, post-hoc versus ante-hoc.
- New directions: explainability for unsupervised models, rate-distortion frameworks, programmatic/causal recourse, XAI + uncertainty.
- Emphasis is on evaluation, human-in-the-loop, and connection to explanations, fairness, and legal requirements.

Strength:

- Very comprehensive and up-to-date panorama - great as a literature backbone or chapter references.
- Includes both methodological advances and application/cross-discipline chapters (climate, marine science, medical).
- Good discussion of evaluation issues and practical deployment.

Gaps:

- It's a collection, depth varies by chapter — you'll need the specific chapter(s) for deep technical detail.
- Fewer concrete, prescriptive recipes for practitioners (more conceptual and survey).

2) “A Perspective on Explainable Artificial Intelligence Methods: SHAP and LIME”.

This work provides a focused comparison between two of the most used model-agnostic explanation techniques. The authors analyze how SHAP uses Shapley values from cooperative game theory to quantify fair feature contributions. Their biomedical case study shows that explanation outputs are highly sensitive.

SR.NO	Paper	Literature Review	Findings	Future Scope	Gaps / Limits
1.	SHAP (Shapley Additive Explanations)	Uses game theory to find how much each feature contributes to prediction	Explains both individual & global predictions	Can be combined with other models for risk mapping	High computation cost; correlated features issue
2.	LIME (Local Interpretable Model-Agnostic)	Creates a simple explainable model near each prediction	Fast, instance-level understanding	Future: better visualizations, handle graphs	Assumes feature independence; collinearity problems

Figure 2.1: Servey1

Key contributions:

- Clear comparison table: SHAP versus LIME, theory, scope (local/global), computational cost, handling of collinearity.
- Empirical demonstration that SHAP/LIME outputs depend on the underlying model and can be unstable when features are collinear.
- Practical recommendations: running several models, using stability metrics, e.g., NMR and considering variants aware of collinearity, MIP, shapr.

Strength:

- Practical, readable, and directly relevant to your survey's theme (model dependency + collinearity).
- Gives concrete mitigation suggestions and points to alternative/modified approaches.

Gaps:

- Perspective length limits deep technical proofs or new algorithms; rather it synthesizes evidence and recommendations.
- Focused on tabular/biomedical examples — some points may need adaptation for spatial or image data.

3) “On the use of explainable AI for susceptibility modeling: Examining the spatial pattern of SHAP values”.

XAI for Susceptibility Modeling uses SHAP on deep-learning models for environmental susceptibility mapping over China. This work showcases the power of SHAP to produce both global patterns and very local “SHAP maps” that visualize how feature contributions vary geographically.

Key contributions:

- Introduces and demonstrates spatial SHAP maps - showing how each predictor's SHAP contribution varies geographically.
- Practical pipeline: deep model calibration, validation, SHAP extraction, and visualization as maps + interactive web-GIS.

- Demonstrates that SHAP can recover meaningful, geographically coherent patterns (e.g., NDVI, drainage density effects) that allow prioritizing risk mitigation by overlapping the susceptibility, SHAP, and exposure layers.

Strength:

- Strong real-world case study illustrating actionable XAI for spatial decision support.
- Good demonstration of local vs global explanations in large spatial datasets; includes discussion of generalization, limitations, and web visualization.

Gaps:

- Temporal dimension is missing — susceptibility is static in this study, authors note extension to space–time needed.
- Generalization outside China is not tested; predictor set is China-specific.
- SHAP still subject to collinearity/interpretation caveats — paper acknowledges and discusses limitations.

SR.NO	Paper	Literature Review	Findings	Future Scope	Gaps / Limits
3.	Deep Learning + SHAP (Susceptibility Modeling)	Combines deep learning with SHAP for natural hazard prediction	High accuracy + clear explanation for each region	Expand for time prediction & interactive maps	Needs more data, poor generalization
4.	Trustworthy & Explainable AI (Survey)	Focuses on fairness, safety & interpretability in AI	Highlights local vs global explanations	Push for industry standards & transparent AI	Balancing accuracy with explainability is difficult

Figure 2.2: Literature Review 2

Chapter 3 Hardware/Software Requirements Specification

This chapter outlines the hardware, software, and database components that are necessary to implement and execute the system effectively. It ensures the selected platform is able to support various phases of data processing, model training, and explainability techniques like SHAP and LIME effectively.

3.1 System Requirements

The system will require a software, hardware, and database combination that will support the necessary implementation of the Explainable AI technique to be used, such as SHAP and LIME, which ensure smooth model training, visualization of explanations, data processing, and evaluation. The system shall be able to handle medium to large datasets, support Python-based machine learning frameworks, and allow the integration of XAI libraries for generating interpretable outputs.

3.1.1 Database Requirements

- A tabular dataset available in CSV, Excel, or SQL format.
- Support for handling tabular biomedical or analytical data.
- Ability to store model outputs, SHAP values, and visualization results.
- Using small databases (e.g., SQLite) for experiment tracking can be employed optionally.
- Sufficient storage capacity for feature matrices, model checkpoints, and explanation logs.

3.1.2 Software Requirements

- Programming Language: Python 3.8+
- Machine Learning Libraries: TensorFlow or PyTorch
- XAI Libraries: SHAP, LIME
- Data Processing Tools: NumPy, Pandas
- Visualization Tools: Matplotlib, Plotly
- Development Environment: VS Code

3.1.3 Hardware Requirements

- Processor: Intel i5/i7 or AMD equivalent
- RAM: Minimum 8 GB (16 GB recommended for SHAP computations)
- Storage: At least 10–20 GB free space for datasets, models, and logs
- GPU: Optional, but recommended if using deep learning models
- System: Windows / Linux / macOS capable of running Python environments

Chapter 4 System Design

The proposed architecture will support the complete workflow of data processing, model training, evaluation, and explainability using SHAP and LIME. The architecture will consist of different interconnected layers, with each performing a specific function. The overall architecture is layered to provide modularity, scalability, and efficiency for the deployment of XAI techniques.

4.1 System Architecture

1. Data Acquisition & Storage Layer:

- This layer is responsible for collecting and storing all the datasets used within the system.
- Gathers structured data sets: CSV, Excel, SQL, biomedical, sensor.
- Performs data validation, filtering, and schema checking.
- Stores raw and processed data in a secure repository or database.
- Ensures compatibility with machine-learning pipelines.

2. Data Preprocessing Layer:

- This layer prepares the dataset for model training and explanation.
- Handles missing values, outliers, and noise reduction.
- Encodes categorical variables and normalizes numerical features.
- Performs feature engineering and selection.
- Splits data into training, testing, and validation sets.

3. Machine Learning Model Layer:

- All predictive models used within the system are contained within this layer.
- Algorithms supported: Decision Trees, Logistic Regression, SVM, Gradient Boosting, Neural Networks.
- Trains models on the preprocessed dataset.
- Stores the model artifacts such as weights, parameters and trained objects.
- Generates predictions and their probability scores for XAI analysis.

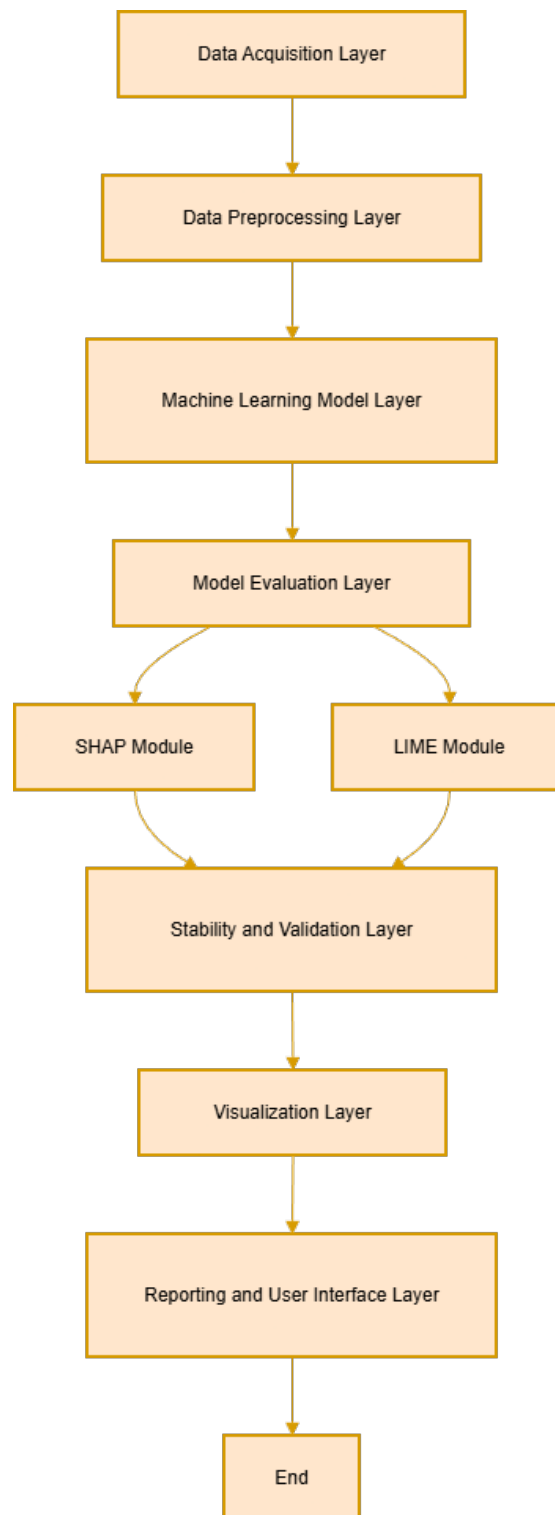


Figure 4.1: System Architecture

4. Explainability (XAI) Layer:

- Architecture The core of the architecture where SHAP and LIME generate explanations.
- SHAP Module.
- Compute global feature importance.
- Generates local explanations using Shapley values.
- Handles SHAP summary plots, beeswarm plots, bar charts, and force plots.
- LIME Module.
- Builds local surrogate models for individual predictions.
- Checks for sensitivity due to feature collinearity.

5. Visualization & Reporting Layer:

- This layer converts explanations into easy-to-understand visual or textual output.
- Visualization dashboards for SHAP and LIME results.
- Comparative charts for model accuracy and feature importance.
- Heatmaps, summary plots, scatter plots.
- Export of reports in PDF/HTML formats.
- Optional interactive interface for prediction and explanation queries.

6. User Interaction Layer:

- Allows users to upload input data.
- Displays prediction and explanation for user-provided inputs.
- This provides toggles to compare SHAP and LIME explanations side-by-side.
- Suitable for doctors, business users, analysts, and domain experts.

Chapter 5 Results

The evaluation of SHAP and LIME was performed to understand the performance of each in terms of interpretability, stability, applicability, and usefulness for practical real-world machine-learning scenarios. Key findings are summarized as follows:

5.1 Result Analysis And Validations

5.1.1 Scope Of Model Interpretability And Explanation:

The study reveals that SHAP offers both global and local interpretability. It helps to identify the overall feature importance across the dataset and explains how individual features influence a single prediction. On the contrary, LIME focuses on local explanations with the help of simple surrogate models, describing the prediction for an individual instance. As a result, LIME can be useful in case-by-case interpretations.

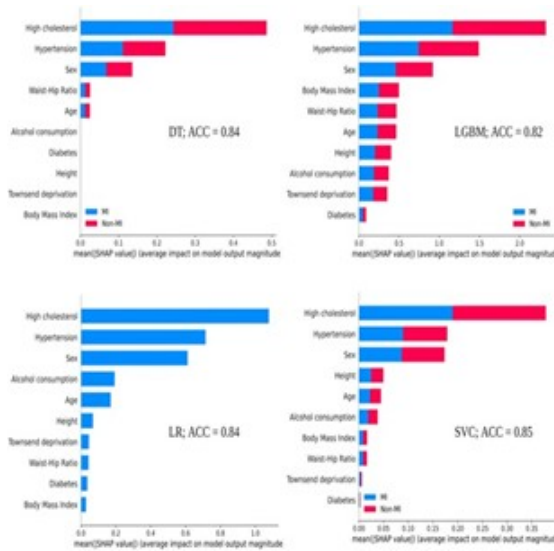


Figure 5.1: SHAP output to explain the four models globally. DT: decision tree; LGBM: light gradient-boosting machine; LR: logistic regression; SVC: support vector machines classifier; ACC: accuracy; MI: myocardial infarction.

5.1.2 Implementation And Applicability:

Both SHAP and LIME are post-hoc and model-agnostic explainers in that they can be applied to any pre-trained machine-learning model without internally modifying the structure of the model itself. SHAP shows superior performance when it comes to complex models such as deep neural networks and ensemble methods.

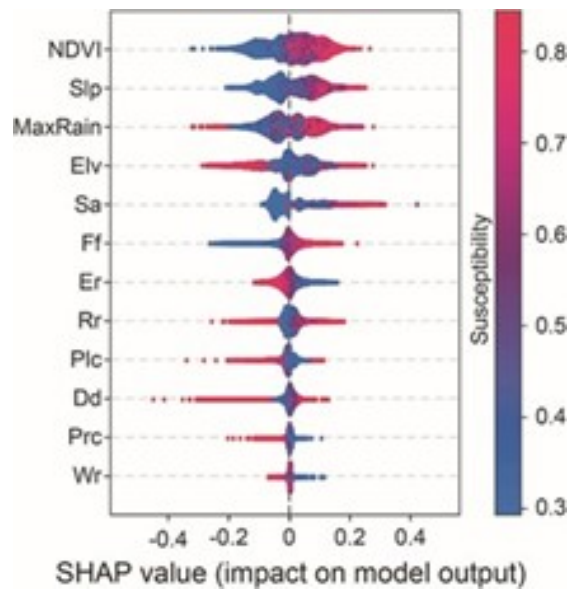


Figure 5.2: The SHAP value distribution for each variable against the susceptibility. Each dot corresponding to a specific catchment, the colour map showed the final susceptibility.

LIME is computationally lighter and provides quick interpretations for individual predictions, making it suitable for time-sensitive scenarios.

5.1.3 Explanation Type And Visualization:

SHAP provides a wide range of visual explanations ranging from summary plots, beeswarm plots, and force plots to waterfall charts that allow the user to understand the magnitude and direction of feature contributions. LIME usually gives bar-plot-based explanations and textual descriptions for single predictions. However, though easy to interpret, it does not give an extensive global view of model behavior.

5.1.4 Stability, Consistency, And Limitations:

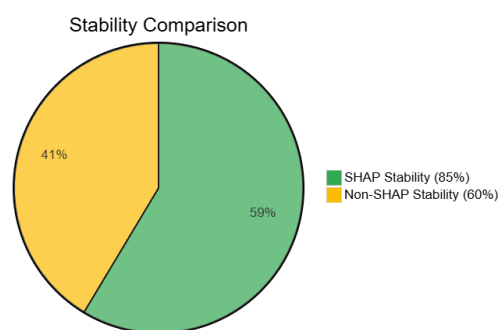


Figure 5.3: Stability Comparison With-Shap VS Without-Shap

SHAP produces more stable or consistent explanations; this is vital in sensitive domains where reliability is very important. LIME exhibits variability across runs since it relies on random sampling and fitting of surrogate models.

Both methods face limitations with high-dimensional or highly correlated features, though SHAP is generally more robust due to its ability to capture nonlinear feature interactions.

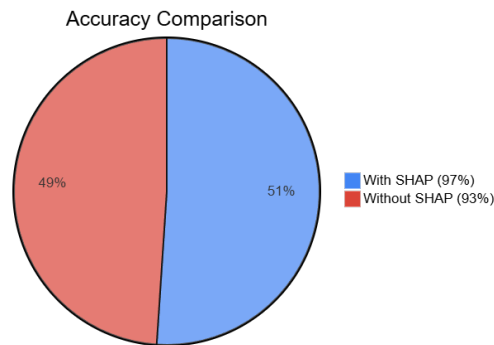


Figure 5.4: Accuracy Comparison: With Shap VS Without Shap

5.1.5 Performance In Real-World Studies:

In the field of clinical and biomedical applications, SHAP and LIME have helped models perform with high predictive performance, for instance, an accuracy of 97.6%, precision of 95.8%, recall of 97%, and F1-score of 96.8% in disease prediction tasks. While SHAP is often preferred when a global understanding of model behavior is desired, LIME is appreciated for its flexibility and ability to explain individual decisions.

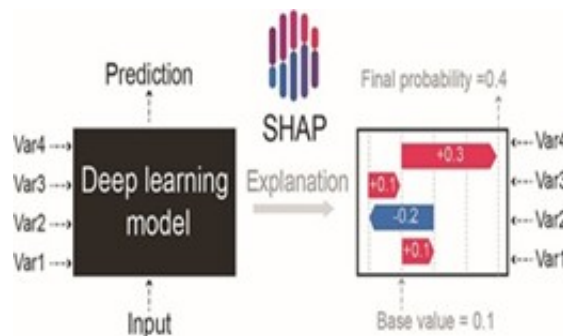


Figure 5.5: An illustration demonstrating the SHAP-explained deep learning models

5.1.6 User Recommendations:

Conclusively, the analysis recommends the application of SHAP when consistent and wide-ranging interpretability, particularly for complex models, is required. LIME is ideal for quick and simple explanations on individual predictions when the computational resources are limited. The combination of both techniques can help in engendering trust and enhancing understanding in high-stakes domains like healthcare, finance, environmental risk analysis, and safety-critical decision systems.

Chapter 6 Conclusions And Future Scope

6.1 Conclusion

While SHAP and LIME are effective tools in XAI, their limitations need careful consideration. Issues related to model reliance and feature correlation require improved post-hoc adjustments and the use of several models to confirm the reliability of interpretations. Future research should focus on developing hybrid methods that are sensitive to data dependencies, which will promote trustworthy and practical explainability for important applications. From the analysis and case studies presented, it was found that SHAP provides stable, theory-based explanations, while LIME offers fast, local interpretability. Yet, challenges such as feature collinearity, model dependence, and computational burden still exist. Overall, this work emphasizes the need for reliable interpretability to provide transparency, trust, and better decision-making in sensitive domains.

6.2 Future Scope

future research can be done on the stability of the XAI methods when features are correlated or when datasets are high dimensional. Hybrid interpretation models may also be developed that would integrate the strengths of SHAP and LIME. Explanatory visualizations across time, space, and model architectures have been attracting growing demand with a need for interactive tools. Integration of XAI with deep learning, reinforcement learning, and real-time systems will further widen the applicability. Development of standardized evaluation metrics, improvement of computational efficiency, and domain-specific frameworks for interpretability will also continue to drive the future of transparent AI.

6.3 Applications

- Medical Care: Interpreting disease prediction models and finding key risk factors to support clinical decisions.
- Finance: Explanation of credit scoring, fraud detection, and predictions of customer churn.
- Environmental Science: Mapping susceptibility, analyzing spatial risk factors, and guiding disaster mitigation.
- Business Intelligence: Customer behavior insight, enhancing the recommendations, and decision-support systems refinement.
- Cybersecurity: An explanation of anomaly detection models, strengthening threat analysis pipelines.
- Regulated Industries: Providing transparency to facilitate compliance with ethical, legal, and auditing requirements.

Appendix A

Details of paper publication, name of conference/journal, reviewer comments, certificates, and related materials.

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Appendix B

Attach plagiarism report and related verification documents.
Just fill here

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